```
#include <iostream>
#include <fstream>
#include <string>
using namespace std;

struct employee {
   string name;
   int emp_id;
} emp;
struct student {
   int ID;
   string name;
   int year;
   string department;
   string reason;
   string dormitory_sign;
   string library_sign;
   string cafteria_sign;
   string registrar_sign;
} stud;
void main_menu();
void dormitory();
void library();
void cafteria();
void registrar();
void student();
void employee();
void registerStudent();
void updateClearanceStatus();
void displayStudents();
void searchStudentById();
void uploadstud_info();
int dormi_id = 1220,cafe_id =12001,libr_id = 31301,regis_id = 43131;
int ch;
int main(){
    system("color 06");

cout<<"@@@
```

```
        cout<<"@@@ 5.chot tut----------4163/16 @@@\n";

cout<<"@@@
```

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        cout << "2. Search Student by ID\n";
        cout << "3. Back to main menu\n";
        cout << "0. Exit\n";
        cout << "Enter your choice: ";
        cin >> ch;
        system("cls");
        switch (ch) {
            case 1:
                registerStudent();
                break;
            case 2:
                searchStudentById();
                break;
            case 3:
                main_menu();
                break;
            case 0:
                exit(0);
            default:
                cout << "Invalid choice! Please try again.\n";
        }
    } while (ch<4);
}
void dormitory_menu(){
            system("color CE");
       cout << "Enter dormitory ID: ";
       cin >> emp.emp_id;
     if (emp.emp_id ==dormi_id) {
    do{
    cout<<"Wellcome to dormiotry staff:\n";
    cout << "1. Show all student status\n";
                cout << "2. Update student status\n";
                cout << "3. Back to employee menu\n";
                cout << "0. Exit\n";
                c:
                cin>>ch;
                system("cls");
                switch(ch)
                {
                 case 1:
                  displayStudents();
                  break;
                 case 2:
                  dormitory();
                  break;
                 case 3:
                  employee();
                  break;
```

```
            case 0:
             exit(0);
            default:
             cout<<"Invlid choice! please try again\n";
             goto c;
   }
          }while(ch<4);
        }
         else{
          cout<<"Dormitory id not found\n";
   }
}
void cafterian_menu(){
    system("color cE");
    cout << "Enter cafterian ID: \n";
    cin >> emp.emp_id;
     if (emp.emp_id == cafe_id) {
     do{
    cout<<"Wellcome to cafterian staff:\n";
  cout << "1. Show all student status\n";
         cout << "2. Update student status\n";
         cout << "3. Back to employee menu\n";
         cout << "0. Exit\n";
         v:
         cin>>ch;
         system("cls");
         switch(ch){
          case 1:
            displayStudents();
            break;
          case 2:
            cafteria();
            break;
          case 3:
            employee();
            break;
          case 0:
            exit(0);
          default:
           cout<<"Invlid choice! please try again\n";
          goto v;
   }
          }while(ch<4);
        }
         else{
          cout<<"Cafterian id not found\n";
   }
}
```

```
void liberary_menu(){
        system("color cE");
            cout << "Enter librarian ID: \n";
       cin >> emp.emp_id;
     if (emp.emp_id = libr_id) {
    do{
    cout<<"Wellcome to librarian staff:\n";
    cout << "1. Show all student status\n";
               cout << "2. Update student status\n";
               cout << "0. Exit\n";
               x:
               cin>>ch;
               system("cls");
               switch(ch)
               {
                case 1:
                 displayStudents();
                 break;
                case 2:
                  library();
                 break;
                case 0:
                 exit(0);
                default:
                 cout<<"Invlid choice! please try again\n";
                 goto x;
    }
   }while(ch<4);
             }
            else{
             cout<<"Librarian id not found\n";
    }
}
void registrar_menu(){
        system("color cE");
             cout << "Enter registrar ID: \n";
       cin >> emp.emp_id;
     if (emp.emp_id == regis_id) {
     do{
    cout<<"Wellcome to registrar staff:\n";
    cout << "1. Show all student status\n";
               cout << "2. Update student status\n";
               cout << "0. Exit\n";
               q:
               cin>>ch;
               system("cls");
               switch(ch)
               {
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                case 1:
                 displayStudents();
                 break;
                case 2:
                 registrar();
                 break;
                case 0:
                 exit(0);
                default:
                 cout<<"Invlid choice! please try again\n";
                 goto q;
    }
   }while(ch<4);
             }
            else{
             cout<<"Registrar id not found\n";
    }

}
void employee() {
    system("color EA");
    do {
     cout<<"WELLCOME TO EMPLOYEE INTERFACE \n";
        cout << "1. dormitory_menu\n";
        cout << "2. liberary_menu\n";
        cout << "3. cafterian_menu\n";
        cout << "4. registrar_menu\n";
        cout << "5. Back to main menu\n";
        cout << "0. Exit\n";
        cout << "Enter your choice: ";
        cin >> ch;
        system("cls");
        switch (ch) {
            case 1:
                dormitory_menu();
                break;
            case 2:
             liberary_menu();
                break;
            case 3:
             liberary_menu();
                break;
            case 4:
             registrar_menu();
                break;
            case 5:
                main_menu();
                break;
```

```
            case 0:
               exit(0);
            default:
               cout << "Invalid choice! Please try again.\n";
         }
      } while (ch < 6);


}

void registerStudent() {
   system("color cE");
   ofstream file("mtu_clearance.txt", ios::app);
   cout << "Enter ID: ";
   cin >> stud.ID;
   cout << "Enter Name: ";
   cin >> stud.name;
   cout << "Enter Department: ";
   cin >> stud.department;
   cout << "Enter Year of Study: ";
   cin >> stud.year;
   cout << "Enter Reason: ";
   cin >> stud.reason;
   system("cls");
   stud.dormitory_sign = " pending ";
   stud.library_sign = " pending ";
   stud.cafteria_sign = " pending ";
   stud.registrar_sign = " pending ";
   cout<<"successfully registered\n";
   file<< stud.ID << "\t" << stud.name << "\t\t" << stud.department << "\t\t" <<
stud.year << "\t" << stud.reason << "\t\t" << stud.dormitory_sign  << "\t\t" <<
stud.library_sign << "\t\t" << stud.cafteria_sign<< "\t\t" << stud.registrar_sign
<< "\n";
   file.close();
}

void displayStudents() {
   system("color AE");
   ifstream file("mtu_clearance.txt");
   if(file.is_open()){
   file.seekg(0,ios::beg);

cout<<"stud_ID\tStud_Name\tDepartment\tYear\tReason\t\tdormitory\tlibrary
\t\tcafteria\tRegistrar"<<endl;
   while (file >> stud.ID >> stud.name >> stud.department >> stud.year >>
stud.reason >>  stud.dormitory_sign >>
stud.library_sign>>stud.cafteria_sign>>stud.registrar_sign ) {
       cout<< stud.ID << "\t" << stud.name << "\t\t" << stud.department << "\t\t"
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<< stud.year << "\t" << stud.reason << "\t\t" << stud.dormitory_sign  << "\t\t" <<
stud.library_sign << "\t\t" << stud.cafteria_sign<< "\t\t" << stud.registrar_sign
<< "\n";
    }
    file.close();
}
else{
 cout<<"File is doesn't exist!\n";
}
}
void dormitory(){
    system("color AB");
 int id;
    cout << "Enter Student ID to update: ";
    cin >> id;
    ifstream file("mtu_clearance.txt");
    ofstream tempFile("temp.txt");
    while (file >> stud.ID >> stud.name >> stud.department >> stud.year >>
stud.reason >>  stud.dormitory_sign >>
stud.library_sign>>stud.cafteria_sign>>stud.registrar_sign ) {
        if (stud.ID == id) {
        cout << stud.ID << "\t" << stud.name << "\t" << stud.department << "\t" <<
stud.year << "\t" << stud.reason << "\t" << stud.dormitory_sign <<"\n";
        if(stud.dormitory_sign=="pending"){
         cout<<"press 1.pure:\n";
         cout<<"press 2.panished:\n";
         cin>>ch;
         switch(ch){
          case 1:
          stud.dormitory_sign="pure";
          cout<<"successfully updated\n";
    break;
    case 2:
    stud.dormitory_sign="panshed";
    cout<<"successfully updated\n";
    break;
   }
         tempFile << stud.ID << "\t" << stud.name << "\t\t" << stud.department <<
"\t\t" << stud.year << "\t\t" << stud.reason << "\t\t" << stud.dormitory_sign  <<
"\t\t" << stud.library_sign << "\t\t" << stud.cafteria_sign<< "\t\t" <<
stud.registrar_sign << "\n";
   }
   else if(stud.dormitory_sign=="pure"){
    cout<<"The student is already pured\n";
   }
   else{
    cout<<"The student is already panished\n";
   }
```

```
        }
    }
    tempFile.close();
 file.close();
 remove("mtu_clearance.txt");
    rename("temp.txt", "mtu_clearance.txt");
}
void library(){
    system("color cE");
 int id;
    cout << "Enter Student ID to update: ";
    cin >> id;
    system("cls");
    ifstream file("mtu_clearance.txt");
    ofstream tempFile("temp.txt");
    while (file >> stud.ID >> stud.name >> stud.department >> stud.year >>
stud.reason >>  stud.dormitory_sign >>
stud.library_sign>>stud.cafteria_sign>>stud.registrar_sign ) {
        if (stud.ID == id) {
        cout << stud.ID << "\t" << stud.name << "\t" << stud.department << "\t" <<
stud.year << "\t" << stud.reason << "\t" <<stud.library_sign<< "\n";
        if(stud.library_sign=="pending"){
         cout<<"press 1.pure:\n";
         cout<<"press 2.panished:\n";
         cin>>ch;
         switch(ch){
          case 1:
          stud.library_sign="pure";
          cout<<"successfully updated\n";
    break;
    case 2:
    stud.library_sign="panshed";
    cout<<"successfully updated\n";
    break;
   }
         tempFile << stud.ID << "\t" << stud.name << "\t\t" << stud.department <<
"\t\t" << stud.year << "\t\t" << stud.reason << "\t\t" << stud.dormitory_sign  <<
"\t\t" << stud.library_sign << "\t\t" << stud.cafteria_sign<< "\t\t" <<
stud.registrar_sign << "\n";
   }
   else if(stud.library_sign=="pure"){
    cout<<"The student is already puered\n";
   }
   else{
    cout<<"The student is already panished\n";
   }
        }
    }
```

```
   tempFile.close();
 file.close();
 remove("mtu_clearance.txt");
   rename("temp.txt","mtu_clearance.txt");
}
void cafteria(){
   system("color cE");
 int id;
   cout << "Enter Student ID to update: ";
   cin >> id;
   system("cls");
   ifstream file("mtu_clearance.txt");
   ofstream tempFile("temp.txt");
   while (file >> stud.ID >> stud.name >> stud.department >> stud.year >>
stud.reason >>  stud.dormitory_sign >>
stud.library_sign>>stud.cafteria_sign>>stud.registrar_sign ) {
       if (stud.ID == id) {
       cout << stud.ID << "\t" << stud.name << "\t" << stud.department << "\t" <<
stud.year << "\t" << stud.reason <<"\t" << stud.cafteria_sign << "\n";
       if(stud.cafteria_sign=="pending"){
        cout<<"press 1.pure:\n";
        cout<<"press 2.panished:\n";
        cin>>ch;
        switch(ch){
         case 1:
         stud.cafteria_sign="pure";
         cout<<"successfully updated\n";
   break;
   case 2:
   stud.cafteria_sign="panshed";
   cout<<"successfully updated\n";
   break;
  }
        tempFile << stud.ID << "\t" << stud.name << "\t\t" << stud.department <<
"\t\t" << stud.year << "\t\t" << stud.reason << "\t\t" << stud.dormitory_sign  <<
"\t\t" << stud.library_sign << "\t\t" << stud.cafteria_sign<< "\t\t" <<
stud.registrar_sign << "\n";
  }
  else if(stud.library_sign=="pure"){
   cout<<"The student is already pured\n";
  }
  else{
   cout<<"The student is already panished\n";
  }
      }
      else{
       cout<<"no student is registered by this Id\n";
 }
```

```
    }
    tempFile.close();
 file.close();
 remove("mtu_clearance.txt");
    rename("temp.txt", "mtu_clearance.txt");

}
void registrar(){
    system("color cE");
 int id;
    cout << "Enter Student ID to update: ";
    cin >> id;
    system("cls");
    ifstream file("mtu_clearance.txt");
    ofstream tempFile("temp.txt");
    while (file >> stud.ID >> stud.name >> stud.department >> stud.year >>
stud.reason >>  stud.dormitory_sign >>
stud.library_sign>>stud.cafteria_sign>>stud.registrar_sign ) {
        if (stud.ID == id) {
        cout<< stud.ID << "\t" << stud.name << "\t\t" << stud.department << "\t\t"
<< stud.year << "\t" << stud.reason << "\t\t" << stud.dormitory_sign  << "\t\t" <<
stud.library_sign << "\t\t" << stud.cafteria_sign<< "\t\t" << stud.registrar_sign
<< "\n";
        if(stud.dormitory_sign=="pure" && stud.library_sign=="pure" &&
stud.cafteria_sign=="pure" && stud.registrar_sign=="pending"){
        cout<<"press 1.to cleared:\n";
        a:
        cin>>ch;
  if(ch==1){
  stud.registrar_sign="cleared";
  cout<<"successfully updated\n";
      }
      else{
       cout<<"please enrter only 1 to cleared student";
       goto a;
   }
     }
 else if (stud.dormitory_sign=="panshed"||stud.library_sign=="panshed"||
stud.cafteria_sign=="panshed"&&stud.registrar_sign=="pending"){
  cout<<"The student have punishement\n";
  cout<<"press 1.to uncleared:\n";
        w:
        cin>>ch;
  if(ch==1){
  stud.registrar_sign="uncleared";
  cout<<"successfully updated\n";
      }
      else{
```

```
      cout<<"please enter only 1 to uncleared student";
      goto w;
   }

     }
     else if (stud.registrar_sign=="cleared"){
      cout<<"the student already cleared\n";
   }
   else if (stud.registrar_sign=="uncleared"){
      cout<<"the student already uncleared\n";
   }
     else{
   cout<<"You haven't finished in other department! Please first finish at all
department\n";
     }
        tempFile << stud.ID << "\t" << stud.name << "\t\t" << stud.department <<
"\t\t" << stud.year << "\t\t" << stud.reason << "\t\t" << stud.dormitory_sign  <<
"\t\t" << stud.library_sign << "\t\t" << stud.cafteria_sign<< "\t\t" <<
stud.registrar_sign << "\n";
   }
   else{
    cout<<"Student id is not found\n";
   }
   }
         tempFile.close();
 file.close();
 remove("mtu_clearance.txt");
    rename("temp.txt", "mtu_clearance.txt");
    }
void searchStudentById() {
    system("color cE");
    int id;
    cout << "Enter Student ID: ";
    cin >> id;
  system("cls");
    ifstream file("mtu_clearance.txt");
    while (file >> stud.ID >> stud.name >> stud.department >> stud.year >>
stud.reason >>  stud.dormitory_sign >>
stud.library_sign>>stud.cafteria_sign>>stud.registrar_sign ) {
      if (stud.ID == id) {

cout<<"stud_ID\tStud_Name\tDepartment\tYear\tReason\t\tdormitory\tlibrary
\t\tcafteria\tRegistrar"<<endl;
       cout<< stud.ID << "\t" << stud.name << "\t\t" << stud.department << "\t\t"
<< stud.year << "\t" << stud.reason << "\t\t" << stud.dormitory_sign  << "\t\t" <<
stud.library_sign << "\t\t" << stud.cafteria_sign<< "\t\t" << stud.registrar_sign
<< "\n";
    }
```

```
    }
    cout << "Student ID not found.\n";
    file.close();
}
```